

Streaming Live RAG

Start retrieving while the user is still talking — and keep every answer tied to the corpus.

Team member	What they built
Yash	Architecture, two-stage controller, async pipeline, packaging
Yuvraj Bansal	Hybrid Qdrant search, GPU embeddings, reranker
Aditya Gupta	Live VS Arena, voice streaming, waveform UI
Jaanya Bagdi	Grounding validator, session memory, gates G1–G6

People speak requests, not queries

One sentence, several questions

“Pune hall capacity for 30, and the cancellation terms, and catering?” A single embedding blurs these together and the later asks get lost.

Dead air after every turn

Classic RAG waits for silence, then searches, then generates. Our naive baseline leaves about 2.5 s before the first token.

Late details force a restart

“What if we cancel 5 days before?” should sharpen the last answer, not throw away the session and search again.

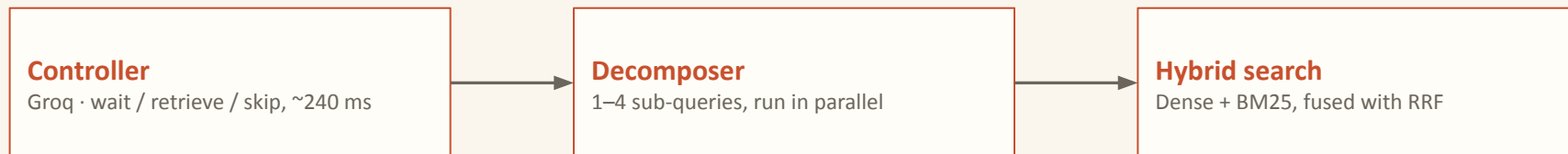
Fast models invent sources

Quick generation can cite a [Doc_99] that never existed. Without a check, an enterprise answer can't be trusted.

Naive: speech ends → ~2.5 s of silence → first token. **Ours:** search already done when speech ends → first token in ~310 ms.

Retrieve while speaking, answer at once

WHILE THE USER SPEAKS (t_i)



WHEN SPEECH ENDS (t_{end})



Shortcuts that save time and tokens

Chit-chat Answered directly, zero searches.

“Make that 2 bullets” Skips retrieval, reformats the last answer.

Unsupported claim One retry, then it says it can’t verify.

A lean stack, no agent framework

Layer	Choice
Transport	FastAPI + asyncio, WebSockets
Controller LLM	Groq · LLaMA 3.3 70B Versatile
Synthesizer LLM	Google Gemini 2.5 Flash
Vector store	Qdrant hybrid, embedded fallback
Dense embeddings	bge-small-en-v1.5, FastEmbed ONNX (CUDA)
Sparse search	Qdrant BM25 + RRF
Reranker	ms-marco-MiniLM-L-6-v2
Grounding	Regex validator, no LLM calls

Why no LangChain / LangGraph

ADR-1

- Nothing in them handles retrieving on a half-spoken sentence.
- Generic graph layers add tens of ms we can't spend.
- Plain asyncio lets us cancel and prune speculative branches.

Dev tools we used (Antigravity/Gemini, Copilot, ChatGPT, Claude) are listed in our AI Disclosure. Everything above is what runs at inference time.

What we did differently

- 1 Speculative retrieval**
Search fires on a stable prefix mid-sentence; a stop-word guard stops it jumping on half-words.
- 2 Two-stage controller**
A fast LLM decides when to fire. Heuristics alone false-triggered on 21–29% of chit-chat; ours: 0%.
- 3 Refine, don't restart**
NEW_TOPIC, LATE_DETAIL or PRESENTATION_ONLY. Late details add delta citations and bump the version.
- 4 Grounding without an LLM**
A short regex check matches every [Doc \$Section] to retrieved chunk IDs. Fabricated IDs never get through.
- 5 Honest thoughts + live arena**
The 4-step thought stream shows real timestamps, no artificial delays. The VS Arena runs both pipelines side by side.

Measured, not claimed

7.9×

faster P50 first token (310 vs 2,450 ms)

9.8 ms

hybrid retrieval on GPU (166.7 ms on CPU)

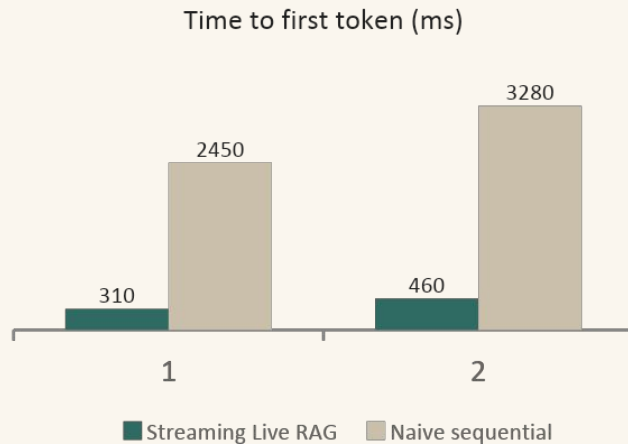
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median cost per turn

0

fabricated citations

Gate	Criterion	Result — all pass
G1	Reproducibility	5/5 checks
G2	Early retrieval	100% early, 0% false trigger
G3	Multi-intent	100% (6/6)
G4	Grounding	100% support, 0 fabricated
G5	Session refinement	10/10 checks
G6	Telemetry	100% (25/25)



RTX 5060 laptop GPU · 173-chunk corpus · 64 labelled queries · arena runs use identical models with cache bypassed.

What we'd be upfront about

Limits today

- Small corpus: 6 documents, 173 chunks, and a labelled set we wrote ourselves.
- On that set hybrid search tied dense-only on recall; we kept it for exact numbers and section IDs.
- Sub-10 ms retrieval needs a GPU. CPU fallback works but takes ~167 ms.
- Controller and synthesizer are cloud APIs, so rate limits and network time are real.
- Voice is browser speech recognition or simulated transcripts.

Where it goes next

- On-device streaming ASR, no network hop.
- Barge-in: cancel in-flight searches when the user cuts in.
- Run embeddings and the controller on Galaxy NPU hardware.
- Knowledge-graph fusion for multi-hop questions.

Thank you — questions welcome